Matteo Benassi

FULL STACK DEVELOPER

Computer Engineering Student · Full Stack & Systems Developer

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EDUCATION

B.Sc. in Computer Engineering | UNIMORE – University of Modena and Reggio Emilia — **2024 – Present (Exp. 2027)**

125/180 ECTS completed · **Weighted GPA: 27.3/30** (2 Honors) · Fully on track with all exams (Mantova campus). Core 3rd-year coursework: *Industrial Informatics* (C/C++, manual memory allocation, POSIX multithreading, Makefiles), *Intelligent IoT* (Python OOP, Smart Factory modeling with uv), *Cyber-physical Security* (network defense, applied cryptography).

High School Diploma in Computer Science & Telecom | ITIS "E. Fermi" Mantova — **Final Grade: 100/100 · 2019 – 2024**

Graduated with **highest honors (100/100)**. Rigorous foundation in software engineering (C, C++, Java, web), algorithms, relational databases, and computer systems architecture.

HACKATHONS & AWARDS

1st Place – Hack4Water 2026 | *Garda Chiese Alert* — **■ 1st Place Winner**

Geolocated public water infrastructure reporting platform built for Consorzio Garda Chiese (33 municipalities, 76,000+ ha). Citizen voice reports parsed via Gemini LLM, strict PostGIS geofencing, and live operator triage dashboard with instant push alerts.

Stack: Django REST Framework · PostgreSQL/PostGIS · React · TypeScript · Leaflet · Gemini AI · Docker

2nd Place – SCIoTeM 2026 | *Smart Beam* — **■ 2nd Place Podium**

Predictive maintenance system for municipal public lighting with interactive GIS mapping, failure risk scoring, and technician ticketing.

Stack: Django · Python · scikit-learn · GIS Mapping

2nd Place – SCIoTeM 2025 | *P7M AI Summarizer* — **■ 2nd Place Podium**

Strips cryptographic PKCS#7 (.p7m) envelopes from digitally signed administrative files and generates keyword-focused semantic summaries with exact page reference citations.

Stack: Django · asn1crypto · Google Gemini API

SELECTED TECHNICAL PROJECTS & SYSTEMS

Home Server & Storage (TrueNAS) | Personal Linux Infrastructure — **Home Server / Self-Hosting**

Home server built on TrueNAS for network-attached storage (NAS), redundant drive management, local backup automation, file sharing, and self-hosted services within a Linux environment.

STM32 Smart Garden (Embedded IoT Firmware) | STMicroelectronics B-L475E-IOT01A — **Embedded C Firmware**

Autonomous smart irrigation system: embedded HTTP web server for settings, Wi-Fi fetch of Open-Meteo forecasts to preemptively conserve water when rain is forecast, and GPIO relay control for solenoid valves.

SyncForge | Python Video File Management Utility — **Open Source Utility**

Custom automation tool engineered in Python to streamline local video file organization, synchronization, and media processing workflows.

Sagre in Lombardia (benpie) | Python & Geospatial Web — **Web & Geospatial**

Interactive geospatial platform displaying cultural events and festivals across the Lombardy region with map visualization.

WORK EXPERIENCE

IT Support Intern | Novellini S.p.A. (Mantova, Italy) — **Jun 2023 – Jul 2023**

Supported corporate IT operations: hardware diagnostics and workstation setup, network infrastructure troubleshooting, and internal user helpdesk support across manufacturing facilities.

TECHNICAL SKILLS & LANGUAGES

Languages:	Python, C, C++, SQL, TypeScript, JavaScript, Java, Bash
Backend & Web:	Django, Django REST Framework, RESTful APIs, Docker, React, Leaflet
Systems, Server & OS:	Linux, TrueNAS (Home Server, storage NAS), POSIX Threads (PThreads), STM32 (C), macOS
Databases & Data:	PostgreSQL, PostGIS, scikit-learn, Google Gemini API
Languages:	Italian (Native) · **English (B2 Cambridge Certified)**
Target Focus:	Motorsport IT & Engineering Systems (telemetry, edge computing, server/network infrastructure)